

Session 2 · Deck 2.B2

Overlapping Generations: Finite Lives

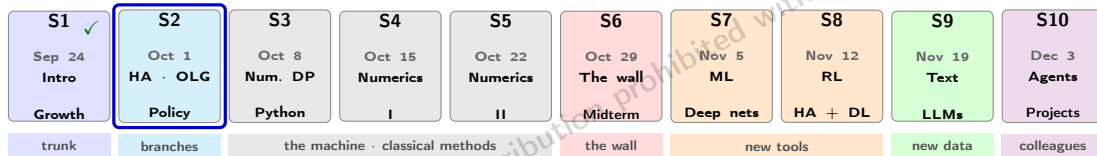
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Build it on paper → solve it on the machine → push it to the frontier with AI

Where We Are

▼ you are here: 2.B2



Build it on paper → solve it on the machine → push it to the frontier with AI

Block B: 2.B1 aggregate risk and the distribution as a state · 2.B2 finite lives and cohorts · 2.B3 the government as a player: Ramsey and time consistency.

A Reform That Is Both a Success and a Failure

Universal health coverage

Chen, Feng & Gu run it in a quantitative life-cycle economy, financed by a consumption tax.

In the aggregate: healthy individuals 86.5% $\rightarrow$ 90.0%; life expectancy at 25 +0.6 **years**; human capital +1.2%.

But not evenly. Life expectancy gains +0.8 **and** +0.9 **years** in income quintiles 1 and 2, and +0.1 in quintile 5 — the rich were already insured. Welfare rises 11.6% of current consumption for the median voter, and the largest gains go to the *middle*, not the poorest, who already had Medicaid.

Chen, Feng & Gu, *IER* 66(1), 2025, 107–141. Steady-state comparison.

Why this needs finite lives

In an infinitely-lived economy there is one dynasty, so “long run” and “the people alive now” are the same agent.

With overlapping generations they are **different people**, and a reform can make one better off by making the other worse off.

Winners from a steady-state comparison can be losers once the transition is counted. Any welfare claim from a life-cycle model that reports only steady states is incomplete.

What Changes When Households Die

- **Age enters the individual state.** Saving behaviour is no longer stationary: the young borrow and accumulate, the old decumulate.
- **The cross-section becomes a mixture** of cohorts, each with its own distribution — not a single stationary pool.
- **New economics appears:** retirement, bequests, survival risk, and intergenerational redistribution through the government.
- **The welfare theorems can fail** for a reason that has nothing to do with incomplete markets — next slide.

Two traditions, one framework

Diamond (1965): two periods, analytically transparent, used to make a theoretical point.

Auerbach–Kotlikoff (1987) and after: 60–80 periods, calibrated, used to evaluate fiscal policy.

We take three slides on Diamond — for one result that infinite-horizon models cannot produce — and then spend the deck on the quantitative version.

Samuelson (1958); Diamond (1965); Auerbach & Kotlikoff (1987).

Diamond in One Slide

Each household lives two periods. Young, it works, earns w_t , consumes and saves; old, it consumes its savings and retires.

$$\max_s \ln c_t^y + \beta \ln c_{t+1}^o$$

subject to $c_t^y = w_t - s_t$ and $c_{t+1}^o = (1 + r_{t+1})s_t$.

With log utility the saving rule is remarkably simple — **independent of the interest rate**:

$$s_t = \frac{\beta}{1 + \beta} w_t.$$

With population growth n , capital per worker evolves as

$$k_{t+1} = \frac{\beta(1 - \alpha)}{(1 + \beta)(1 + n)} k_t^\alpha.$$

Where the saving rate comes from

There *is* an Euler equation here, $1/c_t^y = \beta(1 + r_{t+1})/c_{t+1}^o$ — but it links only the two periods of **one life**. No planner, no dynasty, no transversality condition reaching into a distant future.

With log utility the income and substitution effects cancel, so the rate is $\beta/(1 + \beta)$ whatever r is. **Nothing ties it to what is socially desirable** — the opening for the next two slides.

Note $w_t = (1 - \alpha)k_t^\alpha$ comes from the firm, so the whole economy collapses to a first-order difference equation in k .

Where k^* and k^{gold} Come From

The competitive steady state. The Euler equation gives $s_t = \frac{\beta}{1+\beta} w_t$, the firm gives $w = (1 - \alpha)k^\alpha$, and with full depreciation $1 + r = f'(k) = \alpha k^{\alpha-1}$. Capital per worker is next period's saving per worker, $k_{t+1} = s_t / (1 + n)$, so

$$k_{t+1} = \frac{\beta(1-\alpha)}{(1+\beta)(1+n)} k_t^\alpha, \quad k^* = \left[\frac{\beta(1-\alpha)}{(1+\beta)(1+n)} \right]^{\frac{1}{1-\alpha}}.$$

The golden rule. Maximize steady-state consumption per worker, $c = f(k) - (1 + n)k$:

$$f'(k) = 1 + n, \quad k^{\text{gold}} = \left[\frac{\alpha}{1+n} \right]^{\frac{1}{1-\alpha}}.$$

Compare them

$$k^* > k^{\text{gold}} \iff \frac{\beta(1-\alpha)}{1+\beta} > \alpha$$

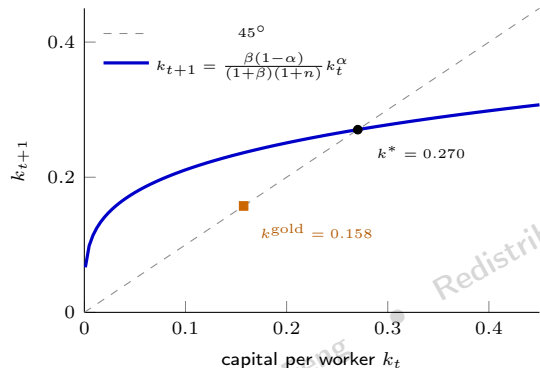
and, equivalently, $f'(k^*) < 1 + n$, i.e.

$$r^* < n.$$

Why they need not agree

k^* is set by **preferences** and the wage share;
 k^{gold} by **technology and demographics**.
Two unrelated objects — nothing in a competitive economy makes them coincide.

A Competitive Equilibrium That Saves Too Much



$\beta = 1$, $\alpha = 0.25$, $n = 0$, full depreciation. Exact closed form;

$k^{\text{gold}} = \alpha^{1/(1-\alpha)}$.

Dynamic inefficiency

Here $k^* > k^{\text{gold}}$. The economy holds *more* capital than the level that maximizes steady-state consumption.

Cut everyone's saving, and consumption rises **today and in every future period**. That is a Pareto improvement — available, and not taken.

The First Welfare Theorem fails. Not because of a missing market or a tax, but because there are **infinitely many agents** and no single budget constraint ties them all together.

The Fix, and Why It Is a Real Policy

Pay-as-you-go social security. Tax the young τ per head, pay it straight to the old.

- It **crowds out private saving**: the young need less capital to fund old age, so k falls toward k^{gold} .
- The implicit return on contributions is the growth rate of the tax base, n (plus productivity growth).
- If $r < n$ — exactly the dynamically inefficient case — that implicit return **beats** the market, and the scheme can make everyone better off.
- If $r > n$, it does not. It just transfers.

The test

Dynamic inefficiency $\iff r^* < n$.

So the case for pay-as-you-go social security is an **empirical** question about rates of return, not a matter of taste.

Beware: in an economy with **uninsurable risk**, safe rates are low for the reasons in 2.A1 — a low r is evidence of precautionary saving, not necessarily of over-accumulation. The comparison must use the right return.

The Quantitative Life-Cycle Household

Households live up to I periods (ages 21 to 100), work until retirement age I_R , and survive from age i to $i + 1$ with probability ϕ_i .

Individual state $s = (i, a, b, e)$:

- i — age
- a — assets
- b — accumulated earnings, which sets the pension
- e — idiosyncratic productivity, an age-dependent Markov chain $\Pi_i(e' | e)$

$$v(s) = \max_{c, h, a'} \left\{ u(c, h) + \beta \phi_i \mathbb{E}[v(s') | s] \right\}$$

with $s' = (i + 1, a', b', e')$ and $v(I + 1) \equiv 0$.

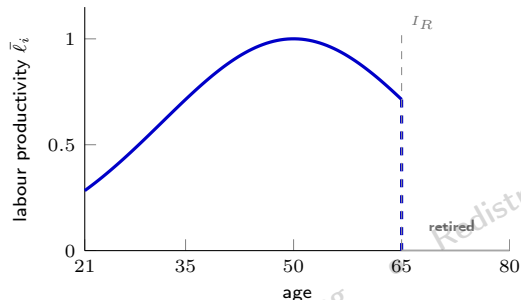
Two discounts, not one

The effective discount factor is $\beta\phi_i$: mortality risk makes older households *behave* more impatiently, with no change in preferences.

b exists only because pensions depend on earnings history — a state variable the **policy** creates.

Finite horizon means **backward induction**: solve age I , then $I - 1$, down to 1 — no fixed point in v .

The Budget Constraint, and Where Policy Enters



Illustrative hump-shaped age-efficiency profile; zero after retirement at I_R .

$$a' = (1+r)a + w e \bar{\ell}_i h$$

$$- \mathcal{T} + \mathcal{B}(i, b) - (1 + \tau_c) c$$

where $\mathcal{T}$ collects income and payroll taxes and $\mathcal{B}(i, b)$ is the pension paid when $i \geq I_R$. **No productivity growth**, so the constraint is in levels.

Three forces shape the wealth profile: the **hump** in earnings, the **cliff** at retirement, and **survival risk**. Together they produce hump-shaped wealth — something the infinite-horizon model cannot generate.

Cohorts: The Distribution Acquires an Index

Let $\lambda(i, a, b, e)$ be the measure of households in state (a, b, e) at age i , with $\sum_{i,a,b,e} \lambda = 1$.

It evolves **forward in age**, not forward in calendar time:

$$\lambda(i+1, a', b', e') = \frac{\phi_i}{1 + \nu} \sum_{a,b,e} \mathbf{1}\{a'(s) = a', b'(s) = b'\} \\ \times \Pi_i(e' | e) \lambda(i, a, b, e)$$

- ϕ_i removes those who die; $1/(1 + \nu)$ rescales for population growth ν .
- The newborn cohort enters with $a = 0$, $b = 0$, and e drawn from Π_1 .

Two fixed points, or one?

Outside, this is still Aiyagari. Prices induce the cohort distributions; the aggregates those imply must reproduce the prices. That outer fixed point in (r, w) is exactly the one from 2.A2.

What is genuinely easier: at *given* prices Aiyagari still needs $\lambda^* = Q^* \lambda^*$ solved in the space of measures. Here only the **age-1** distribution is known and the rest is I applications of a known map. The inner fixed point goes; the outer one stays.

Stationary Equilibrium with Cohorts

Definition

A **stationary recursive competitive equilibrium** consists of prices (r^*, w^*) , value and policy functions (v^*, c^*, h^*, a'^*) defined on $s = (i, a, b, e)$, a measure λ^* , and a fiscal policy, such that:

1. **Household optimality:** given (r^*, w^*) and the policy, (v^*, c^*, h^*, a'^*) solve the life-cycle problem by backward induction from age I .
2. **Consistency of the distribution:** λ^* is generated by a'^* , Π_i , survival ϕ_i and population growth ν , as on the previous slide.
3. **Market clearing:** $K^* = \sum_{i,a,b,e} a \lambda^*$, $L^* = \sum_{i,a,b,e} e \bar{\ell}_i h^*(s) \lambda^*$, with $r^* = F_K - \delta$ and $w^* = F_L$.
4. **The government budget balances:** taxes collected equal spending plus pensions paid.

Definition only — computing it is a nested fixed point, and that is S6.A2.

Add Aggregate Risk and the State Explodes

With an aggregate shock Z , each cohort carries its own distribution and **all of them** are part of the state:

$$S = (Z, \Gamma_1, \Gamma_2, \dots, \Gamma_I),$$

$$K = \sum_i \int a d\Gamma_i, \quad L = \sum_i \int e \bar{\ell}_i d\Gamma_i.$$

The life-cycle Bellman equation becomes

$$V_i(a, e, S) = \max \{u(c, h) + \beta \phi_i \mathbb{E}[V_{i+1}(a', e', S')]\}.$$

Worse than Krusell–Smith

With $I = 60$ cohorts, S contains **sixty** distributions rather than one.

And backward induction no longer rescues us: continuation values now depend on an aggregate state that is itself forecast, so cohorts are coupled through prices.

Where this goes: S6.A2 computes the deterministic transition; S8.B2 handles OLG with aggregate risk using deep learning and a homotopy in the shock variance.

Research in the Wild: Which Equilibrium?

Research in the wild

OLG economies are notorious for **multiple equilibria** and indeterminacy — a difficulty that is structural, not numerical.

Indeterminacy in stochastic overlapping generations models: real effects in the long run, with Matthew Hoelle, *Economic Theory* 63(2), 2017, 559–585.

If a model has several equilibria, a solver returns **one** of them — usually the one nearest your initial guess — and reports no warning. Your policy conclusion can be an artifact of where you started.

What to do about it

- Start from several initial guesses and check you land in the same place.
- Report the range you searched, as with r^* in 2.A3.
- Treat a suspiciously smooth convergence as a question, not a result.

This is the same discipline the course applies to neural solvers in S7–S8. The tool changes; the obligation to verify does not.

Takeaways

- Finite lives put **age** in the state and split the distribution into **cohorts**. The cohort distribution is not a fixed point — it is the newborn distribution pushed forward I times.
- **Dynamic inefficiency** is a genuine failure of the First Welfare Theorem with no friction behind it, and the test is $r^* < n$. It is the theoretical case for pay-as-you-go social security.
- The quantitative model adds survival risk, retirement and a hump-shaped earnings profile — and generates hump-shaped wealth, which no infinite-horizon model does.
- Prices remain a fixed point, and the government budget is an equilibrium condition, not an afterthought.
- Steady-state comparisons can **reverse sign** once the transition is counted. Report both.

Bridge: The Government Has Been Passive So Far

Social security appeared as a **given** rule: a tax rate and a benefit formula, taken as parameters.

But the reform that opened this deck was someone's *choice*. That raises a different question:

- What taxes **should** a government set, given that households respond to them?
- Can it commit to the plan it announces — and what changes if it cannot?
- If policy is a **rule** rather than a path, what does equilibrium even mean?

Next (2.B3)

Ramsey taxation and why past promises become state variables; Chamley–Judd and its fragility; Kydland–Prescott time inconsistency; and Markov-perfect policy as a **fixed point in functions** — the last and hardest consistency condition of the session.